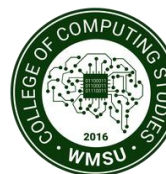




Western Mindanao State University
College of Computing Studies
DEPARTMENT OF INFORMATION TECHNOLOGY
Zamboanga City



CAPSTONE PROJECT AND RESEARCH CAPSULE PROPOSAL

Proposed Research Title

LASINTECH: MFCC-Based Speech Analysis Mobile Application for Real-Time Intoxication Level Classification

Name of the Proponent(s)

Solis, Luisa Mae C.

Research Discipline/Topic

Computer Science / Information Technology; Machine Learning, Audio Signal Processing, Natural Language Processing, Mobile Computing

Sector/Priority Areas/SDG

Sector: Healthcare / Public Safety
Priority Areas: Disease Prevention and Health Promotion; Healthcare Access and Equity
SDG: SDG 3 – Good Health and Well-Being; SDG 11 – Sustainable Cities and Communities

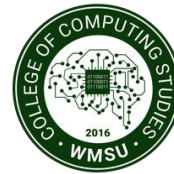
Rationale/Introduction

Alcohol intoxication is a persistent and widespread public safety concern that contributes significantly to road accidents, workplace injuries, physical altercations, and preventable deaths in communities around the world. In the Philippines, the Land Transportation Office and the Philippine National Police have consistently reported alcohol as a primary contributing factor in road crash fatalities. Despite widespread awareness campaigns, many individuals continue to make poor decisions under the influence of alcohol, largely because the physiological effects of intoxication impair the very cognitive faculties needed to recognize one's own compromised state. This creates a critical gap between actual and perceived sobriety that technology can meaningfully help address.

Research in the field of speech processing has consistently demonstrated that alcohol consumption produces measurable and detectable changes in vocal characteristics. Intoxication affects the neuromuscular coordination required for speech production, resulting in observable changes in articulation clarity, speech rate, pitch variability, and pause patterns. Mel-Frequency Cepstral Coefficients (MFCCs) and related acoustic features have been shown in multiple studies to be reliable indicators of these vocal changes, making speech a viable and non-invasive signal for intoxication detection. The proliferation of smartphones equipped with high-quality microphones provides an accessible platform through which this signal can be captured and analyzed.

Despite the availability of breathalyzers and other sobriety tools, no accessible, non-invasive, and hardware-free mobile solution currently exists that enables real-time intoxication classification through speech analysis in a staff-facing context. Existing systems in literature are largely prototype-level or laboratory-based, with limited deployment in real-world mobile environments. Furthermore, most prior studies focus on binary sober/intoxicated classification rather than multi-level severity grading, which is more useful for practical intervention. This gap represents a meaningful opportunity for a capstone study that bridges acoustic signal processing and mobile computing to produce a functional, deployable system.

This study proposes the development of a Machine Learning-Based Speech Analysis Mobile Application that classifies intoxication levels in real time through short audio recordings. The system will extract MFCC and other acoustic features from a subject's speech using Librosa, feed these features into a trained machine learning classifier, and return a categorized intoxication level — Sober, Mildly Impaired, Moderately Impaired, or Severely Impaired — directly on a mobile device. The application is intended to serve as a staff-facing



decision-support tool for use in bars, events, and other venues where alcohol consumption occurs, providing personnel with an objective, technology-driven basis for intervention. The expected output of this study is a fully functional mobile application accompanied by a trained and evaluated ML model, contributing to the body of knowledge on speech-based health and safety systems.

Objectives

General Objective:

To develop an MFCC-Based Speech Analysis Mobile Application capable of classifying intoxication levels in real time through acoustic feature extraction and machine learning classification, and to evaluate its performance and usability as a staff-facing decision-support tool.

Specific Objectives:

- To design and develop a mobile application for real-time audio capture and speech-based intoxication level classification with a user-friendly staff-facing interface.
- To collect, record, preprocess, and annotate audio speech samples from consenting participants representing varying levels of alcohol-induced intoxication.
- To extract and analyze acoustic features — including Mel-Frequency Cepstral Coefficients (MFCCs), pitch, speech rate, and articulation clarity — from collected audio samples using signal processing techniques.
- To implement and train a machine learning classification model capable of categorizing input speech into four intoxication levels: Sober, Mildly Impaired, Moderately Impaired, and Severely Impaired.
- To convert and integrate the trained model into a TensorFlow Lite format suitable for on-device mobile inference without requiring internet connectivity.
- To evaluate the classification model's performance using standard metrics including accuracy, precision, recall, F1-score, and confusion matrix analysis.
- To assess the acceptability and usability of the developed mobile application among target users using a structured evaluation instrument.

Features of the Proposed Study

1. Real-Time Audio Capture Module

- Short audio recording (5–15 seconds) via device microphone
- Automatic noise filtering and audio quality validation
- Waveform preview for staff confirmation before analysis

2. MFCC Feature Extraction Engine

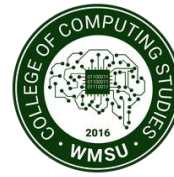
- Extraction of Mel-Frequency Cepstral Coefficients (MFCCs)
- Pitch and speech rate analysis
- Articulation clarity and pause frequency measurement

3. ML-Based Intoxication Classification

- On-device TensorFlow Lite inference
- Output: Sober / Mildly Impaired / Moderately Impaired / Severely Impaired
- Confidence score displayed alongside classification label

4. Classification History and Logs

- Timestamped record of classification results per session
- Summary view of all assessments conducted



5. Privacy-First Architecture

- No raw audio stored or transmitted — only extracted features are processed
- All inference performed locally on the device
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Technologies to be Used:

Mobile Framework: React Native with Expo

Audio Processing: Expo Audio / React Native Audio Recorder

Feature Extraction (Training): Python, Librosa

ML Framework (Training): TensorFlow / Keras, Scikit-learn

On-Device Inference: TensorFlow Lite

Database: SQLite (on-device session logs)

Version Control: GitHub

Development Environment: Visual Studio Code, Google Colab (model training)

Methodology

A. Research and Problem Analysis

The proposed system aims to address the problem of subjective and inconsistent identification of intoxicated individuals in public settings such as bars, restaurants, and events. Specifically, the system will automate the classification of intoxication levels through real-time speech analysis, reducing reliance on human judgment alone. This problem is significant because alcohol impairment is a major contributing factor to road accidents, workplace incidents, and public safety risks, and no accessible mobile tool currently exists for objective, non-invasive staff-side intoxication screening.

B. Research Design and Development Approach

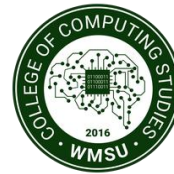
The study will employ a developmental and experimental research design. The developmental aspect covers the design, construction, and iterative refinement of the mobile application and its integrated ML model. The experimental aspect covers the training, testing, and performance evaluation of the classification model. The Agile Software Development Methodology will guide the development process, supporting iterative builds, regular testing, and continuous improvement based on evaluation feedback.

C. Data Requirements and Data Collection

The study requires audio speech samples from participants representing a range of intoxication states. Data will be collected through controlled recording sessions with consenting adult volunteers. Each participant will be asked to read a standardized set of phrases aloud before and at measured intervals after consuming calibrated amounts of alcohol, with BAC levels tracked using a commercial breathalyzer to provide objective ground truth labels. Approximately 500–1,000 labeled audio clips across four intoxication categories will be targeted. Audio data will be preprocessed through noise reduction, silence trimming, resampling to 22,050 Hz, and normalization before feature extraction.

D. Hardware and Software Requirements

Hardware: A smartphone (Android or iOS) will serve as the primary deployment device for the mobile application. A laptop or desktop computer with GPU support will be used for ML model training. A commercially available breathalyzer will be used during data collection to provide BAC-based ground truth labels for audio samples.



Software: Python (Librosa, TensorFlow/Keras, Scikit-learn) for model development; React Native and Expo for mobile app development; TensorFlow Lite for on-device inference; SQLite for local session storage; Visual Studio Code as the primary IDE; GitHub for version control; Google Colab for cloud-based GPU model training. APIs/Third-Party Services: Firebase may be optionally integrated for cloud-based session log synchronization and model update delivery.

E. System Design and Architecture

Input: A 5–15 second audio clip captured via the device microphone.

Process: Audio preprocessing → MFCC and acoustic feature extraction → TensorFlow Lite model inference → intoxication level classification.

Output: Intoxication classification label (Sober / Mildly Impaired / Moderately Impaired / Severely Impaired) with a confidence score displayed on the mobile interface.

Operation Mode: Offline / On-device. No internet connection required for core classification.

User Interaction: Mobile application (Android and iOS via React Native) with a simple, minimal staff-facing interface.

F. Testing and Evaluation

Third-Party Validation: A licensed physician or pharmacologist may be consulted to validate the relationship between observed speech features and medically recognized intoxication indicators.

Testing Methods: Functional testing, accuracy testing, usability testing, performance testing, and User Acceptance Testing (UAT).

Testing Participants: 30–50 respondents from the target user group (bar or venue staff) will participate in usability and acceptance evaluation. Model testing will use a held-out test dataset of approximately 20% of collected audio samples.

Performance Metrics: Model accuracy, precision, recall, F1-score, confusion matrix, inference response time, and user acceptability score.

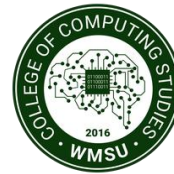
G. Data Analysis and Interpretation

The classification model's performance will be evaluated using accuracy, precision, recall, F1-score, and a confusion matrix to measure per-class and overall classification effectiveness. User acceptance results will be collected via a structured Likert-scale evaluation form and analyzed using weighted mean and descriptive interpretation. Results will be interpreted against acceptable thresholds — a minimum of 80% classification accuracy and a usability score of at least 4.0 out of 5.0 — to determine whether the proposed system meets the defined success criteria.

Expected Output

- A fully functional MFCC-Based Speech Analysis Mobile Application deployable on Android and iOS platforms for real-time intoxication level classification.
- A trained and evaluated machine learning model converted to TensorFlow Lite format for on-device inference.
- A labeled audio speech dataset of intoxication samples that may serve as a foundation for future research.
- A research paper documenting the methodology, model performance, and system evaluation findings, suitable for publication in an academic journal or conference proceedings.

Target Beneficiaries



Bar and Restaurant Staff: The primary users of the system. Staff will benefit from an objective, technology-driven tool that supports decision-making when identifying intoxicated patrons, reducing the social pressure and subjectivity associated with manual assessment.

Event Security Personnel: Security staff at large-scale events and public gatherings can use the application as part of entry screening procedures to identify heavily intoxicated individuals before they pose a risk to others.

Public Health and Safety Organizations: Government agencies and non-governmental organizations focused on road safety and alcohol harm reduction can reference the system as a model for technology-assisted public health intervention.

The Research Community: The labeled dataset and ML methodology produced by this study will contribute to the body of literature on speech-based health monitoring, providing a foundation for future work in intoxication detection, vocal biomarker research, and mobile health systems.

References

- [1] World Health Organization, "Global status report on alcohol and health 2018," WHO. [Online]. Available: <https://www.who.int/publications/i/item/9789241565639>. [Accessed: June 06, 2026].
- [2] P. Milone, M. Costantini, and G. Saggio, "Detection of alcohol intoxication via voice analysis: A systematic review," IEEE Access, vol. 10, pp. 56100–56115, 2022, doi: 10.1109/ACCESS.2022.3177691.
- [3] C. M. Stasak, A. Epps, J. Goecke, and J. Krajewski, "Alcohol intoxication effects on automatic speech processing," in Proc. Interspeech, San Francisco, CA, USA, 2016, pp. 1052–1056.
- [4] J. Krajewski, A. Batliner, and M. Golz, "Acoustic sleepiness detection: Framework and validation of a speech-adapted pattern recognition system," Behavior Research Methods, vol. 41, no. 3, pp. 795–804, 2009, doi: 10.3758/BRM.41.3.795.
- [5] S. B. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," IEEE Trans. Acoustics, Speech, Signal Process., vol. 28, no. 4, pp. 357–366, Aug. 1980, doi: 10.1109/TASSP.1980.1163420.
- [6] B. McFee et al., "librosa: Audio and music signal analysis in Python," in Proc. 14th Python in Science Conf., Austin, TX, USA, 2015, pp. 18–24.
- [7] Department of Health, Philippines, "National Policy on Alcohol," DOH. [Online]. Available: <https://www.doh.gov.ph>. [Accessed: June 06, 2026].
- [8] TensorFlow Developers, TensorFlow Lite, version 2.x. [Online]. Available: <https://www.tensorflow.org/lite>. [Accessed: June 06, 2026].